Univariate Stationary Time Series Models

These notes accompany Lecture 8. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Everything so far has used cross-sectional data. From here on the starting point is different: we exploit the information a variable carries about itself over time. Analysing a single series this way is univariate time series analysis, and it is the subject of this lecture. Multivariate models come later.

Learning outcomes – after this lecture you should be able to:

1. Decompose a series into trend-cycle, seasonal and irregular components, and choose between additive and multiplicative decomposition.
2. Distinguish a stochastic process from the single time series we observe.
3. State the conditions for covariance stationarity and explain why stationarity matters.
4. Write down AR, MA and ARMA processes and define white noise precisely.
5. Use the ACF and PACF to identify the order of a process.
6. Select among competing ARMA models using information criteria and residual diagnostics.
7. Carry out the ARMA modelling procedure on a real series.

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1. Time series data

The obvious characteristic distinguishing time series from cross-sectional data is temporal ordering. Examples used in this lecture include quarterly Polish GDP, daily Bitcoin price changes and monthly CPI in Latvia and Lithuania – different frequencies, but all sharing that property.

1.1 Components of a time series

An observable series can be split into three unobserved components:

Component	What it captures
Trend-cycle (TC)	Long-term evolution over decades – structural change, population growth, technology – together with the smoother medium-term movement from expansion to recession
Seasonal (S)	Fluctuations within the year that repeat regularly, stable in timing, direction and magnitude. Caused by climate, tradition, or administrative habit; may also carry calendar effects. Seasonal effects cancel out over the year as a whole
Irregular (I)	Remaining short-term fluctuations that are neither systematic nor predictable. Transitory and not recurring

1.2 Decomposition

	Additive	Multiplicative
Form	$X_t = TC_t + S_t + I_t$	$X_t = TC_t \times S_t \times I_t$
Seasonal magnitude	Does not change as the trend-cycle level changes	Grows as the trend of the series grows
When used	Required for any series with zero or negative values	Most economic time series behave this way

The seasonally adjusted series is what remains after removing the seasonal component – that is, the combination of trend-cycle and irregular. Note that the irregular component stays in: a seasonally adjusted series is not a smooth series, and smoothness is not a quality criterion.

1.3 Seasonal adjustment software

We use JDemetra+, open source software for seasonal adjustment developed by the National Bank of Belgium in cooperation with the Deutsche Bundesbank and Eurostat, in accordance with European Statistical System guidelines.

It implements the two internationally recommended methods, TRAMO/SEATS and X-13-ARIMA-SEATS, and is officially recommended to members of the ESS and the European System of Central Banks for seasonal and calendar adjustment of official statistics. RJDemetra is the R interface; it relies on rJava and requires Java SE 8 or higher.

Each method has strengths and weaknesses and there is no international consensus on which should be preferred. In practice they often give very similar results. Full operational detail is in the Appendix.

2. Stationarity

2.1 Stochastic process versus time series

A time series is a series of values of some random variable, obtained at successive periods. A sequence of random variables indexed by time is called a stochastic process.

> **The distinction that matters**
>
> Picture a probability distribution sitting at every point in time – a bell curve at t, another at t+1, and so on. The stochastic process is the whole collection of distributions, describing everything that could have happened.
>
> What we actually observe – the dots we plot – is one draw from that process.
>
> We do not know what output growth in Lithuania will be next year; the process describes the possibilities, and the data we eventually collect are one realisation of them.

2.2 Two definitions of stationarity

The key question is whether the probability distribution changes over time. If it does not, the process is stationary; if it does, it is non-stationary.

- **Strict stationarity** – the whole probability distribution is unchanged over time. This is stronger than we need.
- **Covariance (weak) stationarity** – we require only that the first two moments behave:

$$E(Y_t) = E(Y_{t+j}) = \mu \qquad Var(Y_t) = Var(Y_{t+j}) = \sigma^2$$

$$Cov(Y_t, Y_{t+j}) = Cov(Y_s, Y_{s+j}) = \gamma_j$$

In words: the series fluctuates around a constant long-run mean, has a finite constant variance, and the covariance between two observations depends only on the distance j between them, not on where in the sample they sit.

2.3 Why stationarity matters

- If the series is non-stationary, the standard results of classical regression analysis do not apply.
- Regressions involving non-stationary series may be meaningless – the spurious regression problem.
- Shocks to a stationary series are necessarily temporary: their effects dissipate and the series reverts to its long-run mean. Long-horizon forecasts therefore converge on the unconditional mean.

That last point is worth holding on to, because it is exactly what the intercept does in the worked example in Section 4.

3. Identification of time series

3.1 Visual inspection

The first step with any series is to graph it and look. Four generated series illustrate what to look for:

Series	Diagnosis
1	Stationary – an autoregressive process
2	Stationary – a moving average process
3	Non-stationary – the mean drifts. This is trend-stationary: removing the trend makes it stationary
4	Non-stationary – an AR(1) with a structural break. Within each segment the series looks stationary; the level shifts between them

Note what visual inspection cannot do: an AR and an MA process look almost identical in a raw plot. That is why we need the tools in Section 3.3.

3.2 Three processes

Assuming stationarity, three basic types concern us. In each case u_t is a random disturbance and an intercept is allowed.

Autoregressive – y is a function of its own lagged values:

$$y_t = b_0 + b_1 y_{t-1} + \ldots + b_p y_{t-p} + u_t$$

Moving average – y is a function of current and past disturbances:

$$y_t = \mu + u_t + \varphi_1 u_{t-1} + \ldots + \varphi_q u_{t-q}$$

Combined (ARMA) – both kinds of term appear together.

We write AR(p), MA(q) and ARMA(p, q), where p and q are the order – the maximum lag – of each part.

> **A note on notation**
>
> These slides use b for autoregressive coefficients and ϕ (phi) for moving-average coefficients.
>
> Much of the literature uses ϕ for the AR side and θ (theta) for the MA side, so when you read other textbooks, check which convention is in use before comparing formulas.

The disturbance is assumed to be white noise, meaning three things:

$$E(u_t) = 0 \quad Var(u_t) = \sigma^2 \quad Cov(u_t, u_s) = 0 \ \text{ for } t \neq s$$

> **White noise is not the same as i.i.d.**
>
> White noise requires zero mean, constant variance and zero autocovariance. Zero covariance means the observations are uncorrelated – it does not mean they are independent.
>
> An i.i.d. sequence is always white noise; a white noise sequence need not be i.i.d. Two variables can be uncorrelated while still being related non-linearly.

This is not pedantry: the ARCH and GARCH models later in the course are built on exactly this gap – series that are uncorrelated but whose volatility is strongly predictable.

3.3 Autocorrelation and partial autocorrelation

For a series that is not white noise there is dependence between observations. Two tools measure it.

The **autocovariance** at lag j, and the **autocorrelation** obtained by scaling it by the variance:

$$\gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)] \qquad \rho_j = \gamma_j / \gamma_0$$

Plotting ρ_j against j gives the autocorrelation function (ACF), or correlogram.

The **partial autocorrelation** at lag p is the coefficient b_p from a regression of y on all its lags up to p. It measures the relationship between y_t and y_{t-p} after controlling for every intervening lag. For p = 3, regress y_t on y_{t-1}, y_{t-2}, y_{t-3} and read off the third coefficient.

The patterns

Consider an AR(1) with $b_1 = 0.8$ and an MA(1) with $\phi_1 = 0.8$. In the raw plot they are hard to tell apart; in the correlogram they are unmistakable. The AR autocorrelation starts near 0.8 and decays geometrically – 0.8, then 0.8^2, and so on. The MA autocorrelation has one large spike at lag 1 and then drops to essentially zero.

Process	ACF	PACF
AR(p)	Decays gradually toward zero (oscillating if the coefficient is negative)	Cuts off abruptly after lag p
MA(q)	Cuts off abruptly after lag q	Decays gradually toward zero
ARMA(p, q)	Decays gradually	Decays gradually

Practical tips

An ACF that does not go to zero at all may signal non-stationarity – the subject of Lecture 9.

Mixed ARMA processes are genuinely hard to identify, because both functions decay. When both decay gradually, estimate several candidates and compare them.

A workable approach: begin with a parsimonious low-order AR, then inspect the residuals to decide whether MA terms are needed.

4. Estimation and diagnostics

Why fit these models at all? Chiefly for near-term forecasting, which works because macroeconomic and many financial series are persistent.

The workflow: decide on plausible alternative specifications, estimate each, and choose the best on the following criteria.

4.1 Fit versus parsimony

Adding lags automatically improves fit – R^2 rises mechanically – but the added terms may contribute nothing and can degrade forecast quality. We want a small set of significant coefficients, and information criteria formalise the trade-off:

$$AIC = T \cdot ln(SSR) + (p + q + 1) \cdot 2$$

$$SBC = T \cdot ln(SSR) + (p + q + 1) \cdot ln(T)$$

Both are minimised. The Schwarz Bayesian Criterion penalises extra parameters more heavily than Akaike whenever ln(T) > 2, so SBC tends to select more parsimonious models.

4.2 White noise residuals

If the model captures the dynamics properly, nothing systematic should remain in the residuals. Three checks:

- **Plot the standardised residuals** ($\hat{u}_t/\hat{\sigma}$). No more than about 5% should lie outside [−2, +2]. Values beyond that suggest outliers or a structural break.
- **Inspect the residual ACF and PACF.** Autocorrelations should be insignificant at every lag. Remaining autocorrelation indicates the lag structure has not been captured.
- **Run a joint test** on all lags up to s.

The Ljung-Box Q-statistic tests H_o: all $\rho_k = 0$ up to lag s, against the alternative that at least one is non-zero:

$$Q = T(T + 2)\, \Sigma_{k=1..s}\, r_k^2 / (T - k) \quad \sim \quad \chi^2$$

> **Two technical points about Q**
>
> The simpler form $Q = T \cdot \Sigma\, r_k^2$ is the Box-Pierce statistic. Ljung-Box adds the (T+2)/(T−k) weighting, which behaves much better in small samples, and it is what EViews reports.
>
> The degrees of freedom are s minus the number of ARMA parameters estimated, not s. EViews makes this adjustment for you, which is why the first few rows of the correlogram table show no p-value.

4.3 Worked example: euro area inflation

Identification

The plot shows an increase in both the mean and the variance in 2020 – 2022, which hints at non-stationarity, but overall the series looks stationary: inflation returns to the mean, and formal unit root tests (Lecture 9) support treating the series as stationary.

Both the ACF and the PACF decay gradually with no sharp cut-off (PACFs are significant up to lag 4). By the table in Section 3.4 that pattern points to an ARMA rather than a pure AR or MA, so several candidates are worth estimating: AR(2), AR(4) and ARMA(1,1).

Why would inflation be persistent? – the economics behind the statistics

Short-run aggregate supply slopes upward because of nominal rigidity: not every firm adjusts prices and wages immediately. Long-term contracts, menu costs and social norms all delay adjustment.

Expectations and information asymmetry matter too – firms cannot easily tell whether a change in conditions is temporary or permanent, so they respond gradually.

Demand shocks (policy, external demand) and supply shocks (commodity prices, wars, natural disasters, supply-chain disruption) then propagate through this sluggish adjustment, which is exactly what an AR or ARMA structure describes.

Estimation

Model	Key coefficients	AIC	Verdict
AR(2)	AR(1) 0.324, AR(2) 0.227 – both significant at 1%	−0.313	Sound but bettered
AR(4)	AR(3) insignificant (p = 0.37)	–0.353	Over-parameterised
ARMA(1,1)	AR(1) 0.924, MA(1) −0.699 – both significant	−0.369	Lowest AIC and SBC – preferred

Reading the output

- **The intercept is the unconditional mean.** Estimated at 0.18, so euro area monthly inflation fluctuates around a long-run level of about 0.18%. EViews parameterises the model in mean-deviation form, $y_t = \mu + \phi_1(y_{t-1} - \mu) + \ldots$, so use the built-in forecast option rather than computing forecasts by hand.
- **SIGMASQ is the innovation variance.** At 0.0417, the implied shock standard deviation is $\sqrt{0.0417} \approx 0.204$, very close to the reported S.E. of regression of 0.206.
- **Compare that with total volatility.** The standard deviation of inflation itself is 0.23 p.p., against 0.20 for the shocks – so unpredictable shocks account for most of the variation. Plausible sources: energy prices, supply disruptions, policy changes in taxes or spending.

The division of labour between the parts of the model is worth stating explicitly: the intercept defines the anchor, and the AR and MA terms govern the speed and pattern of adjustment back to it. Shocks push inflation away from 0.18; the dynamics bring it back.

Diagnostics

EViews plots the inverse roots of the AR and MA polynomials. Solve the AR polynomial for its roots z and plot 1/z: the process is stationary if all inverse AR roots lie inside the unit circle. A necessary – though not sufficient – condition is that the AR coefficients sum to less than one. Here the ARMA(1,1) passes, so shocks to inflation dissipate rather than exploding.

The inverse MA roots appear on the same chart, but they test something different: invertibility, not stationarity.

The residual correlogram, however, still shows autocorrelation, and the standardised residuals reveal possible outliers. Re-estimating with adjustments improves the fit while leaving the coefficients broadly unchanged – reassuring, because it means the dynamics were not being driven by a handful of unusual observations.

4.4 Stages of ARMA modelling

Stage	What you do
Identification	Examine the time plot, ACF and PACF. Confirm stationarity of mean, variance and covariance, then identify plausible p and q
Estimation	Estimate each tentative model, examine the coefficients, and compare using AIC and SBC – choosing the smallest values. Of the two, SBC is preferable
Diagnostic checking	Assess goodness of fit, plot the residuals, look for outliers and periods of poor fit, and test residual autocorrelation with the Ljung-Box Q-statistic. Avoid overfitting

Appendix. Seasonal adjustment reference

A1. Preparing the data

- **Length.** At least 36 months for monthly data, or 16 quarters for quarterly. Five to seven years is often recommended. But very long series do not automatically give better adjustment – if there are breaks, a shorter span reflecting the current seasonal pattern is better.
- **Chain-linking.** Eurostat suggests adjusting after chain-linking, to avoid artificial seasonality.
- **Outliers.** Many outliers make the seasonal model hard to identify. Four types are distinguished: additive outlier (a single point jump), transitory change (a jump with smooth return), level shift (a permanent change in level), and seasonal break (an abrupt change in the seasonal pattern).

Spreadsheet requirements for JDemetra+: dates in the first column in a format Excel recognises (dd.mm.yyyy); monthly entries as the first day of each month; quarterly entries as the first day of each quarter at three-month intervals; variable names in the first row; numeric data; cell A1 empty; missing values set manually to −99999.

A2. The two methods

	X-13-ARIMA-SEATS	TRAMO/SEATS
Developed by	US Census Bureau	Bank of Spain
Approach	Semi-parametric, based on moving averages	Parametric, based on econometric model estimation
Main drawback / advantage	Uses similar filters for different types of series	Uses data-specific filters

To start: RSAfull in TRAMO/SEATS, or RSA4c / RSA5c in X-13, for series with a significant calendar effect. RSA3 in either method is a good benchmark for a series without calendar effects.

TRAMO/SEATS specifications

Name	Transformation	Calendar effect	Outliers	Modelling
RSA0	Level	None	None	Airline with mean
RSA1	Test for log/level	None	Automatic	Airline with mean
RSA2	Test for log/level	Working day, leap year, Easter	Automatic	Airline with mean

Name	Transformation	Calendar effect	Outliers	Modelling
RSA3	Test for log/level	None	Automatic	Automatic identification
RSA4	Test for log/level	Working day, leap year, Easter	Automatic	Automatic identification
RSA5	Test for log/level	Trading day, leap year, Easter	Automatic	Automatic identification
RSAfull	Test for log/level	Automatic	Automatic	Automatic identification

X-13-ARIMA-SEATS specifications

Name	Transformation	Calendar effect	Outliers	Modelling
X-11	Not applicable	Not applicable	Not applicable	Not applicable
RSA0	Level	None	None	Airline (+ mean)
RSA1	Test for log/level	None	Automatic	Airline (+ mean)
RSA2c	Test for log/level	Working day, leap year, Easter	Automatic	Airline (+ mean)
RSA3	Test for log/level	None	Automatic	Automatic identification
RSA4c	Test for log/level	Working day, leap year, Easter	Automatic	Automatic identification
RSA5c	Test for log/level	Trading day, leap year, Easter	Automatic	Automatic identification

The two tables follow the same logic: RSA0 does no testing at all; RSA1 and RSA2c add automatic outlier detection but keep the Airline model; RSA3 and above switch to automatic model identification. Note that RSA2c uses the Airline model, not automatic identification.

A3. Quality checks

The purpose of quality diagnostics is to reveal weaknesses that would make the results prone to large revisions or to giving false signals about the economy. Modelling problems are more likely with short series containing many outliers.

- Residuals should be independent, random, and normally distributed with zero mean. Check with the residual plots.
- Apply the formal seasonality tests. The combined test checks for stable seasonality across the whole series.
- Check for residual seasonality – over the full sample and over the last three years.
- In JDemetra+, read the Diagnostics summary. "Good" or "uncertain" means the results are acceptable; "severe" means they are not. Remedies: shorten the span to account for breaks, identify outliers explicitly, or include the leap year effect.

Key terms

Term	Meaning
Stochastic process	A sequence of random variables indexed by time; the collection of all possible paths.
Time series	One observed realisation of a stochastic process.
Trend-cycle component	Long-term and medium-term movement, combining structural trend and the business cycle.
Seasonal component	Within-year fluctuations repeating regularly in timing, direction and magnitude.
Irregular component	Short-term fluctuations that are neither systematic nor predictable.
Covariance (weak) stationarity	Constant mean and variance, with covariance depending only on the lag between observations.
Spurious regression	A meaningless regression relationship produced by non-stationary and non-cointegrated series.
White noise	Zero mean, constant variance, zero autocovariance.
AR(p)	y depends on its own p lagged values plus a disturbance.
MA(q)	y depends on the current and q past disturbances.
ARMA(p, q)	Both lagged values of y and lagged disturbances appear.
Autocorrelation function (ACF)	Correlation between observations j periods apart, plotted against j. Also called the correlogram.
Partial autocorrelation (PACF)	The coefficient on lag p in a regression on all lags up to p; the relationship net of intervening lags.
AIC / SBC	Information criteria trading fit against parsimony; minimise. SBC penalises extra parameters more heavily.
Ljung-Box Q-statistic	Joint test that all residual autocorrelations up to lag s are zero.
Box-Jenkins procedure	Identification, estimation, diagnostic checking – iterated until the residuals are white noise.
Inverse roots	Plotted to check stationarity (AR polynomial) and invertibility (MA polynomial); all should lie inside the unit circle.

Review questions

Use these to check your understanding before the seminar.

1. A monthly retail sales series shows seasonal swings that grow as the level of sales grows. Which decomposition would you use, and why would the additive form be wrong here?
2. Explain the difference between a stochastic process and a time series. In what sense do we only ever see one draw?
3. State the three conditions for covariance stationarity. Which of them fails for a series whose variance grows over time?
4. Give two reasons why non-stationarity is a problem for regression analysis.

5. Explain why a shock to a stationary series must be temporary, and what this implies for a long-horizon forecast.
6. Write down an AR(2) and an MA(2). Why are these two nearly impossible to distinguish from a plot of the raw data?
7. You are told a series is uncorrelated at every lag but its volatility is highly predictable. Is it white noise? Is it i.i.d.?
8. The ACF of a series decays geometrically and the PACF cuts off after lag 2. What process is this, and what order?
9. The ACF cuts off after lag 3 and the PACF decays gradually. What process is this?
10. Both the ACF and PACF decay gradually. What does this suggest, and what should you do next?
11. Why can R^2 not be used to choose between an AR(2) and an AR(4)? Which criteria would you use instead?
12. Under what condition does SBC penalise an extra parameter more heavily than AIC? Which tends to select the smaller model?
13. State the null hypothesis of the Ljung-Box test?
14. In the euro area example the intercept is 0.18 and SIGMASQ is 0.0417. Interpret both, and compute the implied shock standard deviation.
15. The standard deviation of inflation is 0.23 and that of the shocks is 0.20. What does the closeness of these two numbers tell you?
16. ARMA(1,1) has a lower AIC and SBC than AR(2) and AR(4). Why is the insignificant AR(3) coefficient in the AR(4) model an argument against that specification too?
17. What do the inverse roots of the AR polynomial test?
18. Your preferred model still shows residual autocorrelation. List three things this could indicate and what you would do about each.
19. List the three stages of the ARMA modelling procedure and say what would send you back from stage three to stage one.

References and further reading

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 13.

JDemetra+ documentation and the ESS Guidelines on Seasonal Adjustment: https://jdemetra-new-documentation.netlify.app.

Macroeconometric Forecasting. IMF online course. URL: https://www.edx.org/course/macroeconometric-forecasting-2.